

Software Design Document (Checkpoint-2)

Smart Classroom Monitoring System for Schedule Compliance & Energy Awareness

1. Document Control

Field	Details
Project Title	Smart Classroom Monitoring System for Schedule Compliance & Energy Awareness
Group ID	FYP-2026-G3
Version	1.0
Date	July 2026
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1. Purpose and Scope

Purpose

This document presents the detailed software design of the Smart Classroom Monitoring System. The purpose of this document is to explain how the requirements identified during Checkpoint-1 are transformed into a practical software design for implementation and testing.

The system uses computer vision techniques to monitor classroom occupancy, compare real-time classroom activities with scheduled timetables, detect unnecessary energy consumption, and generate alerts for academic management.

Scope

The scope of this design includes:

- Classroom occupancy detection using camera feeds.
- Student presence estimation using YOLO object detection.
- Schedule comparison with predefined timetables.
- Detection of lights and fan usage in empty classrooms.
- Alert generation for schedule violations and energy wastage.
- Dashboard and reporting modules.
- Database design and data management.

The following features are outside the scope of this project:

- Student attendance management.
- Face recognition.
- Mobile application support.
- Automatic control of electrical appliances.

2. Reference Documents

Document	Purpose
Topic Approval Form	Confirms approved project title and objectives
Software Requirements Specification (SRS)	Provides system requirements
High-Level Architecture Document	Defines system structure
Project Plan	Defines milestones and tasks
Supervisor Feedback	Provides improvements and corrections

3. Design Goals

Goal	Description
Modularity	System modules are designed independently for easier maintenance.
Accuracy	Occupancy detection should provide reliable results.
Maintainability	Future features can be added without changing core modules.
Usability	Dashboard should be simple and user-friendly.
Security	Only authorized users can access system reports.
Performance	The system should process classroom data in near real-time.
Scalability	Additional classrooms can be added in future versions.

5. System Decomposition

The Smart Classroom Monitoring System is divided into several independent modules. Each module performs a specific function and works together to monitor classroom activities, verify schedule compliance, detect energy wastage, and generate reports.

Module / Component	Purpose	Major Responsibilities	Related Requirement(s)	Owner
Authentication Module	Controls user login and secure system access	Validate user credentials, manage user sessions, and provide role-based access to the dashboard	FR-6	Backend Lead
Camera Capture Module	Captures classroom images for monitoring	Receive video frames from classroom cameras and forward them to the detection module	FR-1	AI/Backend Lead
Occupancy Detection Module	Detects people inside classrooms	Process camera frames using the YOLO model and calculate the number of occupants	FR-1	AI Lead
Schedule Verification Module	Verifies classroom activity with the timetable	Compare occupancy results with scheduled classes and identify delays or missed lectures	FR-2	Backend Lead
Energy Monitoring Module	Detects unnecessary energy consumption	Monitor the status of lights and fans and identify energy wastage in empty classrooms	FR-3	Backend Lead
Alert Management Module	Generates notifications for abnormal conditions	Create alerts for schedule violations, empty classrooms during class hours, and energy wastage	FR-4	Backend Lead
Dashboard Module	Provides monitoring and reporting interface	Display classroom status, occupancy information, alerts, reports, and system statistics	FR-5, FR-6	Frontend Lead
Reporting Module	Generates analytical reports	Prepare daily, weekly, and monthly reports for administrators and coordinators	FR-5	Frontend / Backend Lead
Database Layer	Stores system data	Maintain user information, classroom details, schedules, occupancy logs, energy records, alerts, and reports	FR-1 to FR-6	Database Lead
Activity Logging Module	Records system activities	Store user logins, occupancy detections, generated alerts, report creation, and system events for future analysis	FR-4, FR-5	Backend Lead

6. Detailed Module Design

Module 1: Authentication Module

Field	Module Details
Module Name	Authentication Module
Purpose	Authenticate authorized users and provide secure access to the system dashboard.
Inputs	Email or Username, Password
Outputs	Authenticated user session, assigned user role, or an error message for invalid login credentials
Processing Logic	The system validates the entered credentials against the user database. If authentication is successful, a secure session is created and the user is redirected to the appropriate dashboard according to their role. Invalid login attempts display an error message without revealing system details.
Dependencies	User Database, Session Management, Dashboard Module
Related Requirements	FR-6
Owner	Backend Lead / Frontend Lead

Module 2: Camera Capture Module

Field	Module Details
Module Name	Camera Capture Module
Purpose	Capture real-time classroom images for occupancy monitoring.
Inputs	Camera Feed, Classroom ID
Outputs	Image frames forwarded to the Occupancy Detection Module
Processing Logic	The module continuously captures frames from the classroom camera at predefined intervals and sends them for object detection. It ensures that image quality is sufficient before processing.
Dependencies	Camera Device, Occupancy Detection Module
Related Requirements	FR-1
Owner	AI Lead / Backend Lead

Module 3: Occupancy Detection Module

Field	Module Details
Module Name	Occupancy Detection Module
Purpose	Detect and count people inside classrooms using the YOLO object detection model.
Inputs	Captured Camera Frames
Outputs	Occupancy Count, Detection Status
Processing Logic	The YOLO model processes each captured frame to identify human objects. The system calculates the total number of occupants and stores the detection results in the database for further analysis.
Dependencies	Camera Capture Module, YOLO Model, Database Layer
Related Requirements	FR-1
Owner	AI Lead

Module 4: Schedule Verification Module

Field	Module Details
Module Name	Schedule Verification Module
Purpose	Compare classroom occupancy with the official timetable to determine schedule compliance.
Inputs	Occupancy Count, Classroom ID, Timetable Data
Outputs	Schedule Compliance Status, Violation Report
Processing Logic	The system retrieves the scheduled class for the selected classroom and compares it with the occupancy result. If no occupants are detected during an active class or an occupied room has no scheduled class, the system records a schedule violation.
Dependencies	Occupancy Detection Module, Schedule Database
Related Requirements	FR-2
Owner	Backend Lead

Module 5: Energy Monitoring Module

Field	Module Details
Module Name	Energy Monitoring Module
Purpose	Detect unnecessary energy consumption in classrooms.
Inputs	Occupancy Status, Appliance Status (Lights/Fans)
Outputs	Energy Usage Record, Energy Wastage Alert
Processing Logic	The module checks whether classroom lights or fans remain switched on while the room is empty. If unnecessary energy consumption is detected, the event is stored in the database and an alert is generated.
Dependencies	Occupancy Detection Module, Database Layer
Related Requirements	FR-3
Owner	Backend Lead

Module 6: Alert Management Module

Field	Module Details
Module Name	Alert Management Module
Purpose	Notify administrators and coordinators about important system events.
Inputs	Schedule Violations, Energy Wastage Events
Outputs	Alert Notification, Alert Log
Processing Logic	The module analyzes system events and generates alerts whenever schedule non-compliance or energy wastage is detected. All alerts are recorded for future reporting and analysis.
Dependencies	Schedule Verification Module, Energy Monitoring Module, Database Layer
Related Requirements	FR-4
Owner	Backend Lead

Module 7: Dashboard and Reporting Module

Field	Module Details
Module Name	Dashboard and Reporting Module
Purpose	Display monitoring information and generate analytical reports.
Inputs	Occupancy Logs, Schedule Records, Alerts, Energy Data
Outputs	Dashboard, Charts, Daily Reports, Monthly Reports
Processing Logic	The module retrieves data from the database and presents classroom occupancy, schedule compliance, energy usage, and alert statistics using tables and graphical charts. Users can filter reports based on classroom and date.
Dependencies	Database Layer, Alert Management Module
Related Requirements	FR-5, FR-6
Owner	Frontend Lead / Backend Lead

Module 8: Database Management Module

Field	Module Details
Module Name	Database Management Module
Purpose	Store, organize, and manage all system data.
Inputs	User Information, Timetable Data, Occupancy Logs, Energy Records, Alerts
Outputs	Updated Database Records
Processing Logic	The module performs Create, Read, Update, and Delete (CRUD) operations while maintaining relationships between tables. It ensures data integrity through primary and foreign key constraints.
Dependencies	All System Modules
Related Requirements	FR-1 to FR-6
Owner	Database Lead

7. Database Design

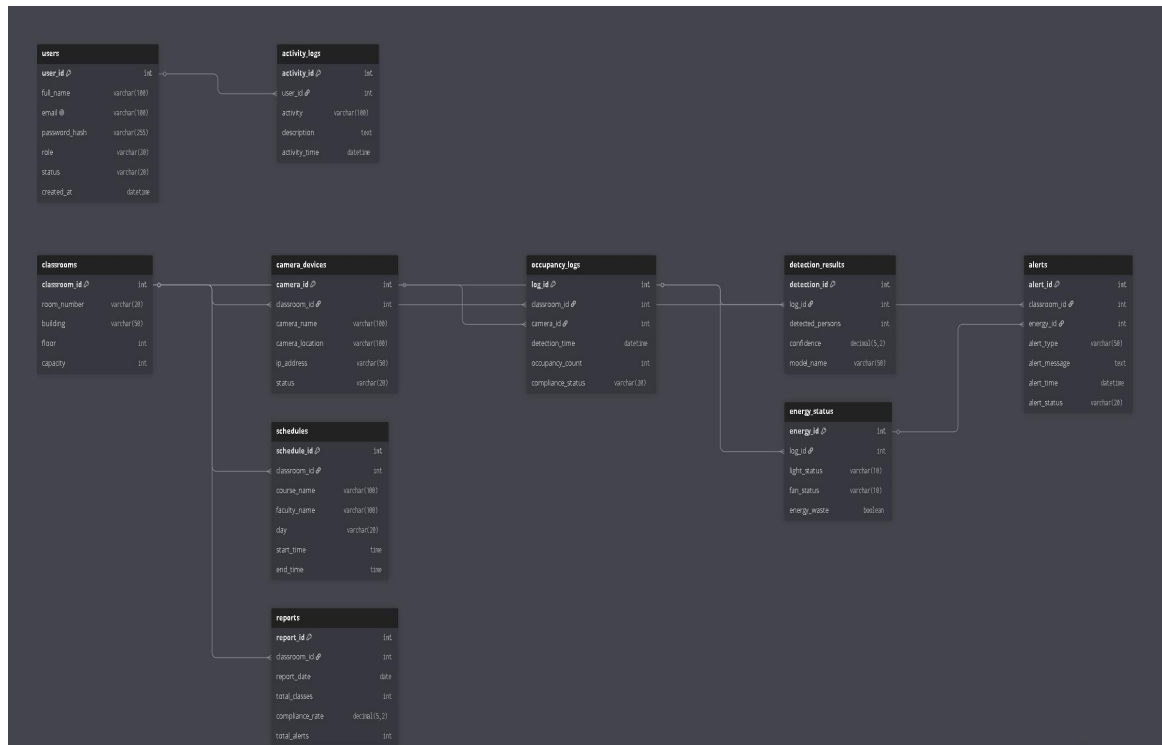


Fig 1 : The key database entities are summarized below.

Table / Entity	Key Attributes	Primary Key	Foreign Keys / Relationships	Important Constraints / Purpose
Users	user_id, full_name, email, password_hash, role, status, created_at	user_id	Referenced by Activity_Logs	Email must be unique; passwords stored as hashes; role determines system access
Classrooms	classroom_id, room_number, building, floor, capacity	classroom_id	Referenced by Camera_Devices, Schedules, Occupancy_Logs, Alerts, Reports	Stores classroom information monitored by the system
Camera_Devices	camera_id, classroom_id, camera_name, camera_location, ip_address, status	camera_id	classroom_id → Classrooms	Stores camera details installed in each classroom
Schedules	schedule_id, classroom_id, course_name, faculty_name, day, start_time, end_time	schedule_id	classroom_id → Classrooms	Defines the official timetable for classrooms
Occupancy_Logs	log_id, classroom_id, camera_id, detection_time, occupancy_count, compliance_status	log_id	classroom_id → Classrooms, camera_id → Camera_Devices	Stores occupancy information detected by the YOLO model
Detection_Results	detection_id, log_id, detected_persons, confidence, model_name	detection_id	log_id → Occupancy_Logs	Stores detailed AI detection results and confidence scores
Energy_Status	energy_id, log_id, light_status, fan_status, energy_waste	energy_id	log_id → Occupancy_Logs	Records classroom energy usage and detects unnecessary energy consumption
Alerts	alert_id, classroom_id, energy_id, alert_type, alert_message, alert_time, alert_status	alert_id	classroom_id → Classrooms, energy_id → Energy_Status	Stores schedule violations and energy wastage alerts
Reports	report_id, classroom_id, report_date, total_classes, compliance_rate, total_alerts	report_id	classroom_id → Classrooms	Generates daily and monthly monitoring reports
Activity_Logs	activity_id, user_id, activity, description, activity_time	activity_id	user_id → Users	Records important user actions for auditing and system monitoring

8. UML / System Models

Diagram / Model	Included?	Purpose / Remarks
Use Case Diagram	Yes	Shows interactions between the Administrator, Academic Coordinator, and the Smart Classroom Monitoring System.
Class Diagram	Yes	Represents the major classes including User, Classroom, CameraDevice, Schedule, OccupancyLog, DetectionResult, EnergyStatus, Alert, Report, and ActivityLog.
Sequence Diagram: Occupancy Detection	Yes	Illustrates the interaction among Camera Device, YOLO Detection Module, Occupancy Detection Module, Database, and Dashboard during classroom monitoring.
Sequence Diagram: Alert Generation	Yes	Demonstrates how occupancy detection and energy monitoring trigger alerts and update the administrator dashboard.
Activity Diagram: Classroom Monitoring	Yes	Describes the complete workflow from image capture to occupancy detection, schedule verification, energy monitoring, and report generation.
Activity Diagram: Energy Monitoring	Yes	Shows the process of checking occupancy, verifying appliance status, detecting energy wastage, and generating alerts.
State Diagram: Classroom Monitoring Status	Yes	Represents classroom state transitions such as Scheduled, Occupied, Class Running, Empty, Energy Wastage, and Alert Generated.
Deployment Diagram	Yes	Shows the deployment of classroom cameras, application server, AI processing module (YOLO), MySQL database, and administrator dashboard.
Wireframes / UI Flow	Yes	Includes Login Screen, Dashboard, Classroom Monitoring Screen, Alerts Screen, Reports Screen, and Schedule Management Screen.

Figure 2: Use Case Diagram of Smart Classroom Monitoring System

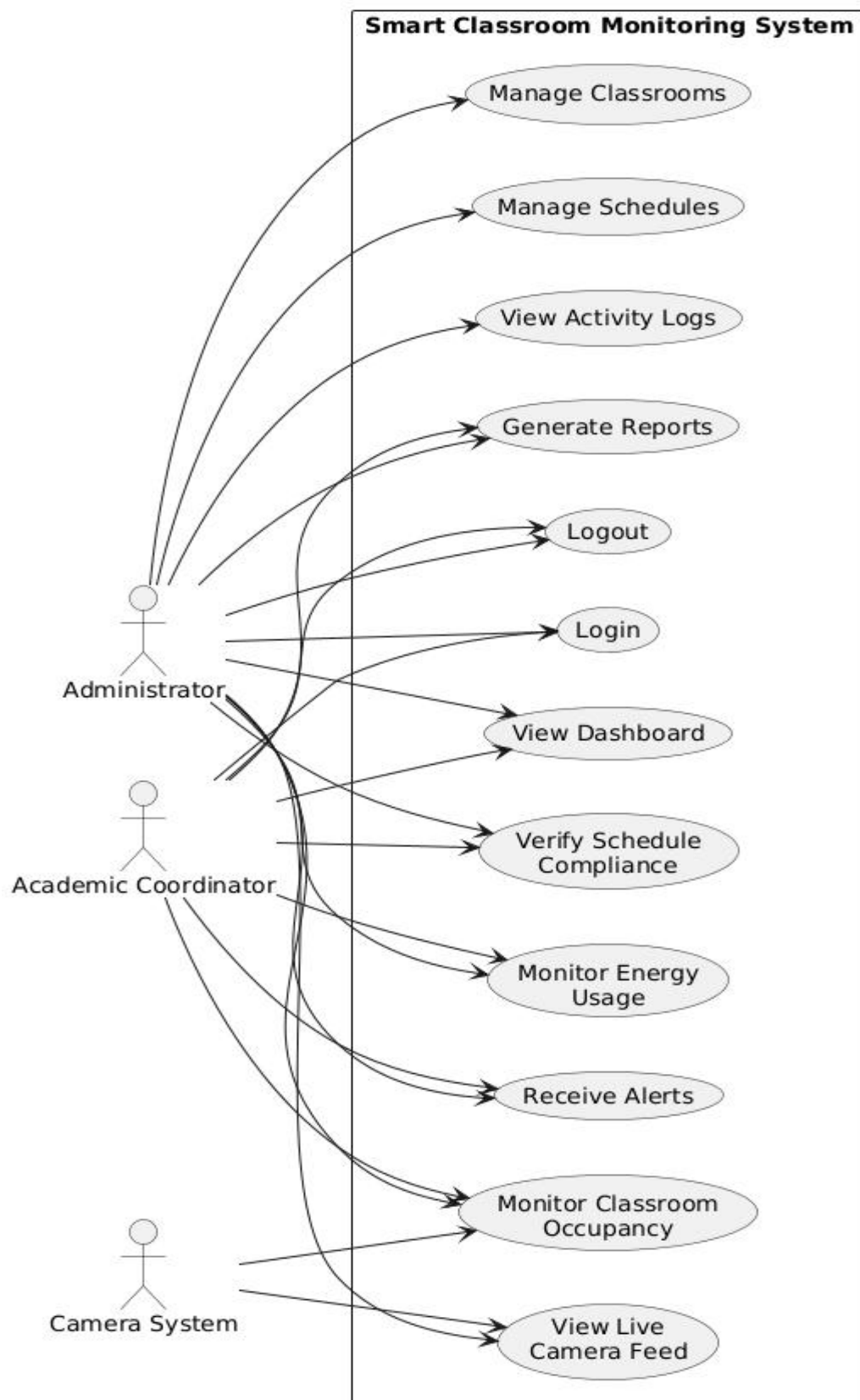


Figure 3: System Context Diagram of Smart Classroom Monitoring System

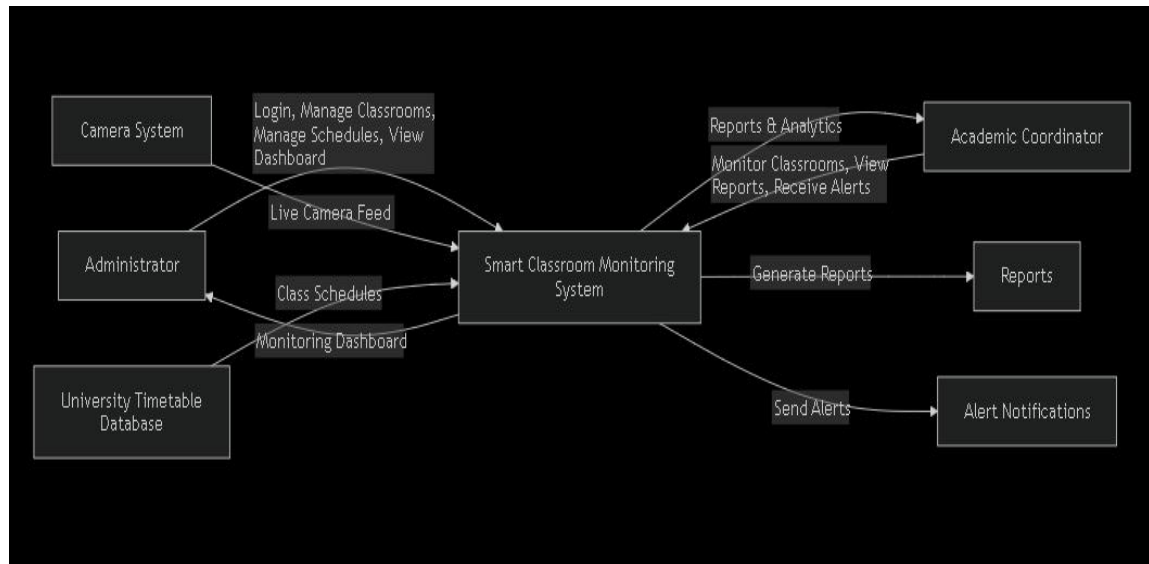


Figure 3 illustrates the System Context Diagram of the Smart Classroom Monitoring System. It shows the interactions between the system and its external entities, including the Administrator, Academic Coordinator, Camera System, and University Timetable Database. The Administrator manages classrooms, schedules, and system settings, while the Academic Coordinator monitors classroom activities, receives alerts, and views reports. The Camera System provides live video feeds for occupancy detection, and the timetable database supplies scheduled class information for schedule compliance verification. The system processes these inputs to generate dashboards, reports, and alert notifications.

Figure 4: Sequence Diagram – Occupancy Detection

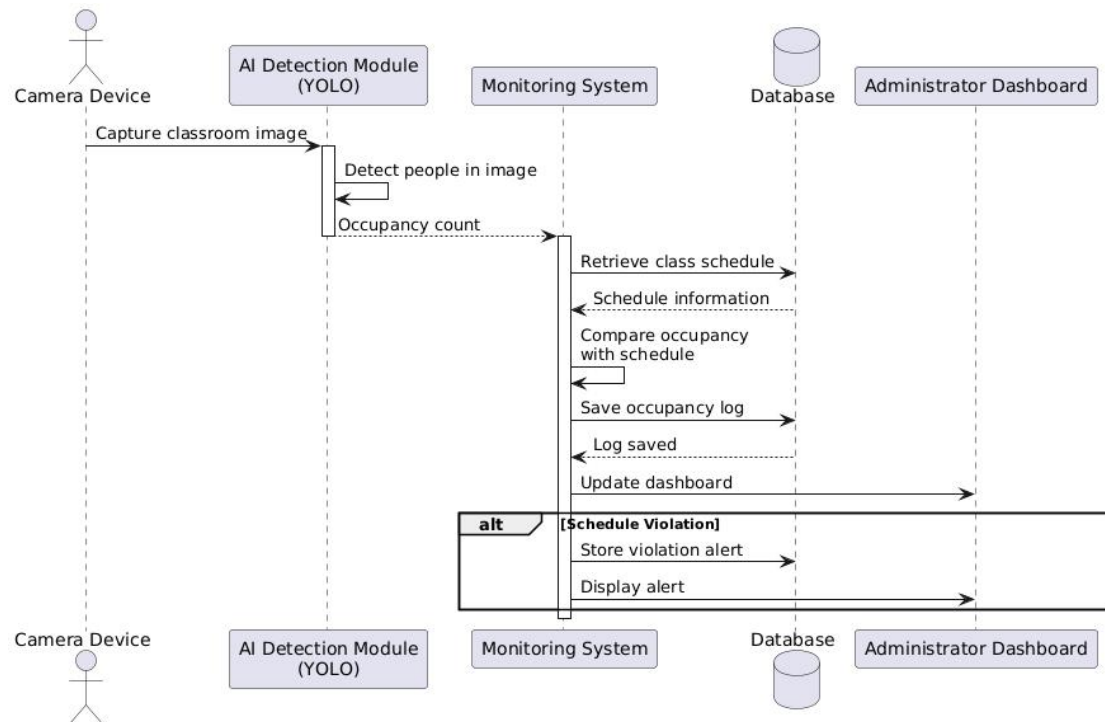
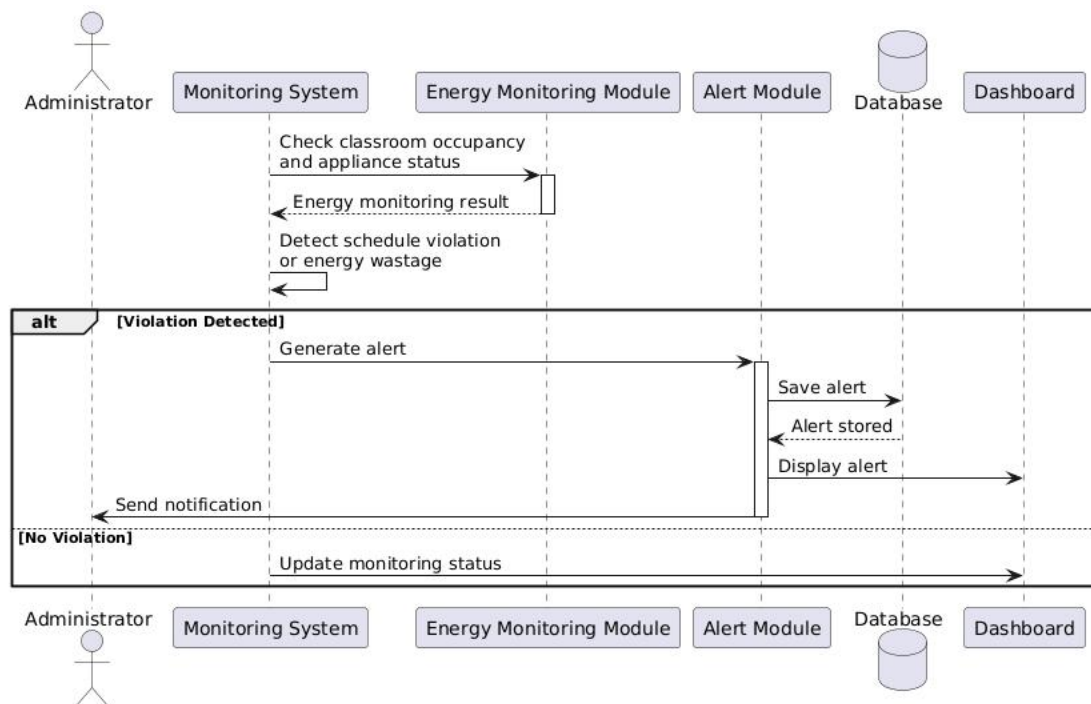


Figure 5: Sequence Diagram – Alert Generation



9. API / Interface Design

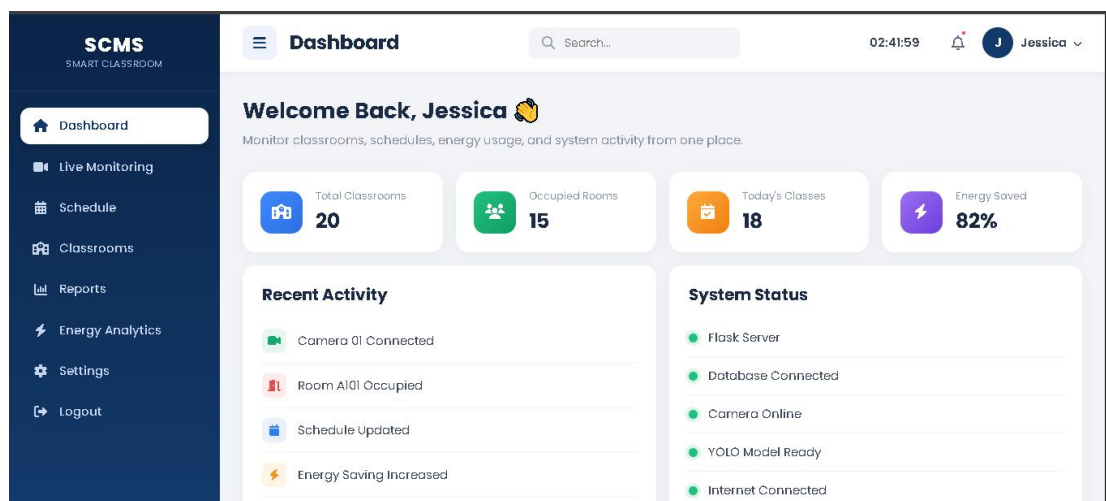
The Smart Classroom Monitoring System follows a modular architecture in which the web dashboard communicates with the backend server through secure application interfaces. These interfaces are responsible for authentication, classroom monitoring, **schedule verification, energy monitoring, alert management, and report generation**. All sensitive operations require authenticated user access to ensure system security and data integrity.

API / Route / Interface	Method	Input	Output	Authentication Required	Related Requirement
/login	POST	Email, Password	User Session / Login Status	No	FR-6
/logout	POST	User Session	Logout Confirmation	Yes	FR-6
/dashboard	GET	User Session	Dashboard Statistics	Yes	FR-5
/classrooms	GET	Classroom ID (Optional)	Classroom Details	Yes	FR-1
/detect-occupancy	POST	Camera Frame, Classroom ID	Occupancy Count	Yes	FR-1
/verify-schedule	POST	Classroom ID	Compliance Status	Yes	FR-2
/energy-monitoring	POST	Occupancy Status, Light/Fan Status	Energy Wastage Status	Yes	FR-3
/alerts	GET	Alert ID	Alert List	Yes	FR-4
/reports	GET	Date Range	Daily / Weekly / Monthly Reports	Yes	FR-5
/activity-log	GET	User ID	Activity History	Yes	FR-6

10. User Interface / Navigation Design

Screen / Page	User Role	Purpose	Main Actions	Related Requirement
Login Page	Administrator	Authenticate users	Login	FR-6
Dashboard	Administrator	View system overview	Monitor statistics	FR-5
Classroom Monitoring	Administrator	Monitor classrooms	View occupancy	FR-1
Schedule Verification	Administrator	Verify timetable	Compare occupancy with schedule	FR-2
Energy Monitoring	Administrator	Detect energy wastage	View light/fan status	FR-3
Alert Management	Administrator	Manage alerts	View & acknowledge alerts	FR-4
Reports	Administrator	Generate reports	Filter & download reports	FR-5
Activity Logs	Administrator	View system activity	Search & review logs	FR-6

Figure 7: Dashboard



11. Security Design Considerations

Security Area	Design Control / Decision	Related Module
Authentication	Only registered users can access the system using a valid email and password. User credentials are verified before granting access.	Authentication Module
Authorization / Role-Based Access	The system implements role-based access control (RBAC). Only administrators are authorized to access monitoring, reporting, and management features.	Authentication Module
Input Validation	All user inputs are validated on both the client and server sides to prevent invalid or malicious data entry.	Authentication Module
Password / Data Protection	User passwords are stored in encrypted (hashed) form within the database to prevent unauthorized disclosure.	Database Management Module
Session Handling	A secure user session is created after successful login and automatically terminated after logout or session timeout.	Authentication Module
Error Handling	Error messages are designed to be user-friendly and do not reveal sensitive system or database information.	All Modules
Database Security	Database access is restricted to authorized backend services, and primary–foreign key constraints ensure data integrity.	Database Management Module
Activity Logging	User logins, system events, alerts, and administrative actions are recorded for auditing and troubleshooting purposes.	Activity Logging Module

12. Error Handling and Validation Design

Error / Validation Case	System Response	User Message / Action	Related Module
Invalid Login Credentials	Authentication request rejected	"Invalid email or password. Please try again."	Authentication Module
Missing Required Fields	Input validation prevents submission	"Please complete all required fields."	Authentication Module
Unauthorized Access	Access denied and user redirected to login page	"You are not authorized to access this page."	Authentication Module
Camera Connection Failure	Monitoring process paused and event logged	"Camera connection failed. Please check the device."	Camera Capture Module
YOLO Detection Failure	Frame processing skipped and next frame requested	"Occupancy detection temporarily unavailable."	Occupancy Detection Module
Missing Timetable Data	Schedule verification skipped	"No timetable is available for the selected classroom."	Schedule Verification Module
Database Connection Failure	Transaction cancelled and error recorded	"Unable to connect to the database. Please try again later."	Database Management Module
Duplicate User Record	User registration/update prevented	"A user with this email already exists."	Database Management Module
Empty Report Request	Report generation cancelled	"No data found for the selected date range."	Reporting Module
Internal System Error	Error logged and administrator notified	"An unexpected error occurred. Please contact the administrator."	All Modules

13. Design Decisions and Trade-offs

Design Decision	Reason / Justification	Alternative Considered	Impact
Use of YOLOv8 for Occupancy Detection	YOLOv8 provides fast and accurate real-time object detection, making it suitable for classroom occupancy monitoring.	Haar Cascade, SSD MobileNet	Improved detection accuracy and faster processing.
Web-Based Dashboard	A web application allows administrators to monitor classrooms from any authorized computer without installing additional software.	Desktop Application	Better accessibility and easier maintenance.
MySQL Relational Database	MySQL provides reliable data storage, supports relational data modeling, and maintains data integrity using primary and foreign keys.	MongoDB	Better support for structured academic data and relationships.
Layered System Architecture	Separates presentation, business logic, AI processing, and database layers to simplify maintenance and future development.	Monolithic Architecture	Improved scalability and modularity.
Role-Based Access Control (RBAC)	Restricts access to sensitive system functions based on user roles.	Single User Access	Enhanced security and better user management.
Alert-Based Monitoring	Administrators are notified immediately when schedule violations or energy wastage are detected.	Manual Monitoring	Faster response time and improved operational efficiency.
Activity Logging	Records important system events for auditing and troubleshooting.	No Logging Mechanism	Easier debugging and accountability.

14. Traceability to Requirements

Requirement ID	Design Element / Module	Diagram / Model	Verification Method
FR-1	Camera Capture Module, Occupancy Detection Module	Use Case Diagram, Sequence Diagram	Functional Testing
FR-2	Schedule Verification Module	Activity Diagram	Scenario-Based Testing
FR-3	Energy Monitoring Module	State Diagram	Functional Testing
FR-4	Alert Management Module	Sequence Diagram	Alert Generation Testing
FR-5	Dashboard and Reporting Module	Wireframes / UI Flow	User Interface Testing
FR-6	Authentication Module	Use Case Diagram	Login and Security Testing

15. Review and Approval

The Software Design Document has been prepared by the project team and reviewed internally before submission. The supervisor will review the document to verify that the proposed design aligns with the approved Software Requirements Specification (SRS), project objectives, and Checkpoint-2 deliverables.

Item	Status / Remarks
Internal Team Review	☒ Yes
Supervisor Remarks	_____
Supervisor Signature	_____
Date	_____